

Gaurav Kumar Gautam

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Research Profile

Geospatial AI and Earth-observation researcher working at the intersection of remote sensing, InSAR, spatial modelling, and machine learning. Research interests include landslide susceptibility and deformation monitoring, spatially rigorous GeoAI, mining-environment assessment, and multi-source satellite analysis. Current work emphasizes physically and geographically meaningful modelling in settings affected by data scarcity and spatial dependence.

Education

Ph.D. in Mining Engineering, Indian Institute of Technology (Indian School of Mines), Dhanbad *Jul 2026 – Present*

Supervisor: Prof. Dheeraj Kumar. Currently completing first-year doctoral coursework; dissertation topic to be defined following the coursework phase.

M.Tech in Geomatic Engineering, Indian Institute of Technology (Indian School of Mines), Dhanbad *Aug 2024 – May 2026*

CGPA: 8.48/10.00

Thesis: Multi-Temporal and Spatial Autocorrelation-Based Assessment of Mining-Induced Ecological Degradation Using a Restoration Priority Index: A Case Study of the Jharia Coalfield, India.

Developed a decadal restoration-prioritization framework for the approximately 370 km² Jharia Coalfield using a ten-year multi-sensor archive. Introduced Vol-NDVI and Surface Alteration Surface (SAS) for mining-landscape assessment and integrated six degradation indicators using Fuzzy AHP. Validation included Moran's $I = 0.8831$ and ROC-AUC = 0.8355.

B.Tech in Civil Engineering, Bihar Engineering University, Patna

Sep 2019 – Sep 2023

CGPA: 7.99/10.00. Final-year project on adapting the GRIHA rating system for an existing institutional campus; Best Final Year Project Award, 2022–23.

Research Experience

Multi-Temporal InSAR Landslide Monitoring, Gangtok, Sikkim

Mar 2026 – Present

Department of Mining Engineering, IIT (ISM) Dhanbad

- Processed 31 ascending and 24 descending Sentinel-1A SLC acquisitions (Jan–Dec 2025) using ISCE2 `topsStack` and `MintPy`, generating 114 ascending and 86 descending interferometric pairs.
- Produced time series containing approximately 6.3 million ascending and 3.7 million descending coherent pixels; LOS velocity ranges are -15.8 to $+41.8$ cm/yr (ASC) and -19.9 to $+10.2$ cm/yr (DSC).
- Joint inversion decomposed LOS observations into east–west (-41.2 to $+18.3$ cm/yr) and vertical (-22.3 to $+25.7$ cm/yr) displacement components; time series captured clear June–September monsoon acceleration.
- Completed all software implementation, InSAR processing, coding, and computational analysis. Manuscript ready for submission; second author/co-author.

GCT-TabDDPM / Synthetic Data for Landslide Inventories

Apr 2026 – Present

Independent methodological research in progress

- Investigating geo-conditional generative modelling for landslide-inventory augmentation, with emphasis on geographic consistency between locations and raster-derived covariates.
- Developing leakage-safe spatial evaluation and auditing failure modes in apparently plausible synthetic geospatial samples.
- Research design, implementation, evaluation framework, and code developed independently. Quantitative results are being finalized; first-author manuscript in preparation.

Jharia Coalfield Restoration Prioritization – M.Tech Thesis

Jan 2025 – Apr 2026

IIT (ISM) Dhanbad; Supervisor: Prof. Dheeraj Kumar

- Built a decadal restoration-priority framework for the approximately 370 km² Jharia Coalfield using Sentinel-1/2, Landsat-8/9, and VIIRS observations.
- Developed two novel indices in the mining context: Vol-NDVI (SAR–optical fusion) and Surface Alteration Surface (SAS), with SAS developed using PCA; weighted six degradation indicators using Fuzzy AHP.
- Applied Theil–Sen trend estimation, Mann–Kendall testing, spatial autocorrelation, and sensitivity analysis. Validation: Moran's $I = 0.8831$; ROC-AUC = 0.8355; Monte Carlo CV = 2.56%; coal-fire capture = 77.48%.
- Carried out the full research workflow, including design, data processing, index development, coding, validation, interpretation, and manuscript preparation. First-author manuscript under review at *Science of Remote Sensing*.

- Knowledge-driven AHP model using 14 geo-environmental factors achieved AUC = 0.844; sensitivity analysis identified slope as the dominant factor.
- Comparative study of AHP, Frequency Ratio, AHP–FR, and Logistic Regression on 211 landslides; AHP–FR achieved success AUC = 0.85 and prediction AUC = 0.81.
- Ensemble study using 677 landslides and five-fold spatial cross-validation: XGBoost AUC = 0.895 and sensitivity = 0.955; WSRF captured 84.5% of landslides within 28.2% of the study area. Partial-dependence analysis identified an instability transition around 25°–30° slope.

Independent Researcher – Hyperspectral Mineral Mapping, Bailadila Iron Ore Mine Summer 2025
Built an EO-1 Hyperion workflow in ENVI using FLAASH atmospheric correction, PCA/MNF reduction, and Spectral Angle Mapper classification to characterize hematite, pyrite, and quartz-associated signatures.

Data Science Intern – TEXMiN, IIT (ISM) Dhanbad Mar 2025 – Aug 2025
Built Python machine-learning workflows for extracting environmental degradation metrics from high-resolution satellite imagery for data-driven mine monitoring.

Contributing Researcher – Groundwater Potential Zonation, Purulia Nov 2024 – Mar 2025
Benchmarked AHP, Frequency Ratio, Logistic Regression, and Random Forest against 66 verified wells in hard-rock aquifers; data-driven models (AUC = 0.75) outperformed expert weighting (AUC = 0.66). Manuscript under review.

Integrated Coastal Flood Risk Assessment, Sagar Island Sep 2024 – Feb 2025
Contributing Researcher
Co-authored a remote-sensing exposure and census-based socioeconomic vulnerability framework for the Indian Sundarbans. Springer book chapter in press.

Publications

Peer-reviewed journal article

- Das, M., **Gautam, G. K.**, Jain, S., Bhat, M. F., Mankar, A. K., & Koner, R. (2026). A comparative analysis of AHP, FR, AHP–FR and LR models for landslide susceptibility mapping in Sikkim Himalaya, India. *Earth Surface Processes and Landforms*, 51(2), e70257.

Manuscripts under review

- **Gautam, G. K.**, Das, M., & Kumar, D. (2026). Multi-temporal and spatial-autocorrelation-based assessment of mining-induced ecological degradation using a Restoration Priority Index: A case study of the Jharia Coalfield, India. *Science of Remote Sensing*. [First author]
- Das, M., **Gautam, G. K.**, & Koner, R. (2025). Benchmarking classical, ensemble, and subspace-based algorithms for landslide susceptibility mapping in the Sikkim Himalaya. *Earth Surface Processes and Landforms*.
- Das, M., **Gautam, G. K.**, & Koner, R. (2025). Modeling landslide susceptibility in the rugged terrain of Sikkim, India. *Geological Journal*.
- Das, M., Dutta, B., Mal, S., Laskar, S., Das, B., & **Gautam, G. K.** (2025). Groundwater potential zonation mapping using deterministic, statistical, and machine-learning approaches: A comparative study of Purulia District, West Bengal, India. *Arabian Journal of Geosciences*.

Book chapter in press

- Patra, S., **Gautam, G. K.**, Das, M., Samanta, D., Jain, S., Bhat, M. F., & Gayen, A. (2025). Integrated coastal flood risk assessment using a multi-criteria GIS approach: A case study of Sagar Island, India. In *Sustainable Solutions*. Springer.

Work in preparation

- **Gautam, G. K.**, et al. (2026). GCT-TabDDPM: A geo-conditional diffusion model for landslide inventory augmentation in the Sikkim Himalaya. [First-author manuscript in preparation]
- Das, M., **Gautam, G. K.**, et al. (2026). Multi-temporal InSAR analysis of landslide deformation in Gangtok, Sikkim, using Sentinel-1A with ascending and descending orbit integration. [Manuscript ready for submission]

Grants and Awards

- **Teaching Assistantship, IIT (ISM) Dhanbad** (2024–2026) – competitive merit award obtained through India's national postgraduate entrance examination; tuition support, stipend, and undergraduate teaching duties.
- **Best Final Year Project Award, Bihar Engineering University** (2022–23).

- **Chief Minister’s Merit-cum-Support Scholarship for Higher Education, Government of Bihar** (2019–2023).

Research Skills

InSAR / SAR	Sentinel-1; ISCE2; MintPy; PSInSAR; SBAS; interferometric time-series processing; multi-geometry decomposition.
Remote sensing	Sentinel-1/2; Landsat-8/9; EO-1 Hyperion; VIIRS; optical, SAR, thermal, nighttime-light, and hyperspectral processing.
GeoAI / ML	XGBoost; Random Forest; WSRF; logistic regression; generative diffusion models; PyTorch; TensorFlow; scikit-learn.
Geospatial computing	Python; GeoPandas; Rasterio; Shapely; NumPy; SciPy; pandas; OpenCV; Google Earth Engine; JavaScript.
GIS / imaging	ArcGIS; QGIS; ENVI; ERDAS Imagine; SNAP.
Spatial statistics	Fuzzy AHP; VIF; Theil–Sen; Mann–Kendall; Moran’s I; Getis–Ord Gi*; Monte Carlo sensitivity analysis; ROC–AUC; spatial cross-validation.
Data	SQL; PostGIS; Power BI; Microsoft Excel.
Languages	English (full professional proficiency); Hindi (native).

Activities and Leadership

- **Chess:** IIT (ISM) General Championship winner; chess team captain; best-player recognition at a Bihar Engineering University inter-college tournament; member, Black Knight Chess Club.
- **Community engagement:** Volunteer teacher with KARTAVYA (2025–present), teaching Physics, Chemistry, and Mathematics to students in Grades 9–12.
- **Sports leadership:** Sports General Secretary, Bihar Engineering University (2022–23); Coordinator, Badminton Club, IIT (ISM) Dhanbad.
- **Field/outdoor experience:** Annapurna Base Camp and Mardi Himal treks; field visits to active coal-mining areas and landslide-affected terrain in Sikkim.